

ISE 5406: Optimization II

Lecture 8

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Reading for this lecture

- Sections 3.3, 3.4 and 3.5 of Bazaraa et al. (2006)
- Section 3.4 of Boyd and Vandenberghe

Outline

Differentiable Convex Functions

Theorem

Let S be a nonempty open convex set in $\mathbb{R}^n$ and $f : S \rightarrow \mathbb{R}$ be differentiable on S . Then f is convex if and only if for any $\bar{\mathbf{x}} \in S$:

$$f(\mathbf{x}) \geq f(\bar{\mathbf{x}}) + \nabla f(\bar{\mathbf{x}})^T (\mathbf{x} - \bar{\mathbf{x}}) \quad \text{for each } \mathbf{x} \in S.$$

Theorem

Let S be a nonempty open convex set in $\mathbb{R}^n$ and $f : S \rightarrow \mathbb{R}$ be twice differentiable on S . Then f is convex if and only if the Hessian matrix $H(\bar{\mathbf{x}}) = \nabla^2 f(\bar{\mathbf{x}})$ is positive semidefinite at each $\bar{\mathbf{x}} \in S$.

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$$[\nabla f(\mathbf{x}_2) - \nabla f(\mathbf{x}_1)]^T (\mathbf{x}_2 - \mathbf{x}_1) \geq 0. \quad (1)$$

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Proof uses the Mean Value Theorem: Let $S \subseteq \mathbb{R}^n$ be a nonempty open convex set and $f : S \rightarrow \mathbb{R}$ be differentiable. Then for each $\mathbf{x}_1, \mathbf{x}_2 \in S$, there exists $\lambda \in (0, 1)$ such that

$$f(\mathbf{x}_2) - f(\mathbf{x}_1) = (\nabla f(\lambda \mathbf{x}_1 + (1 - \lambda) \mathbf{x}_2))^T (\mathbf{x}_2 - \mathbf{x}_1).$$

Characterization of the Set of Optimal Solutions

Note: $\mathbf{x}^* \in \arg \min_{\mathbf{x} \in S} f(\mathbf{x})$ means $\mathbf{x}^*$ is an optimal solution to $\min_{\mathbf{x} \in S} f(\mathbf{x})$.

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$$\arg \min_{\mathbf{x} \in S} f(\mathbf{x}) =: S^* = \{ \mathbf{x} \in S : \nabla f(\mathbf{x}^*)^T (\mathbf{x} - \mathbf{x}^*) \leq 0, \nabla f(\mathbf{x}^*) = \nabla f(\mathbf{x}) \}.$$

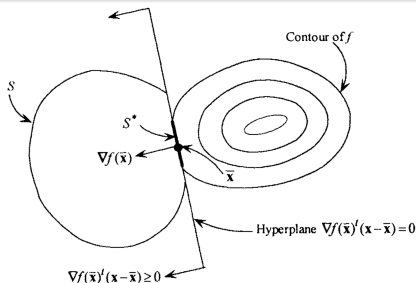
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Corollary 1: $S^* = \{\mathbf{x} \in S : \nabla f(\mathbf{x}^*)^T (\mathbf{x} - \mathbf{x}^*) = 0 \text{ and } \nabla f(\mathbf{x}^*) = \nabla f(\mathbf{x})\}.$

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Corollary 2: If S is polyhedral and f is a convex quadratic function, then S^* is polyhedral.

Maximizing a Convex Function

Theorem: Let S be a nonempty convex set in $\mathbb{R}^n$ and $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be a convex function. If $\mathbf{x}^* \in S$ is a local maximum to $\max_{\mathbf{x} \in S} f(\mathbf{x})$, then

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Note: These conditions are necessary but not sufficient for optimality.

Example: $f(x) = x^2$, $S := \{x \in \mathbb{R} : -1 \leq x \leq 2\}$. Consider $\bar{x} = 0$.

Convex Maximization Has an Extreme Point Solution

Theorem: Let S be a nonempty compact polyhedral set in $\mathbb{R}^n$ and $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be convex. The problem $\max_{\mathbf{x} \in S} f(\mathbf{x})$ has an optimal solution that is an extreme point of S .

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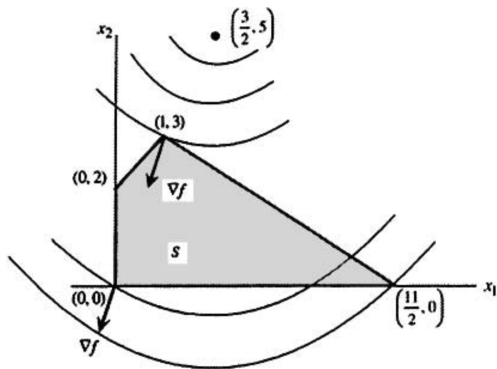
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Can be generalized to non-polyhedral convex sets S .

Example: Let $f(\mathbf{x}) := (x_1 - \frac{3}{2})^2 + (x_2 - 5)^2$ and

$$S := \{\mathbf{x} \in \mathbb{R}_+^2 : -x_1 + x_2 \leq 2, 2x_1 + 3x_2 \leq 11\}.$$



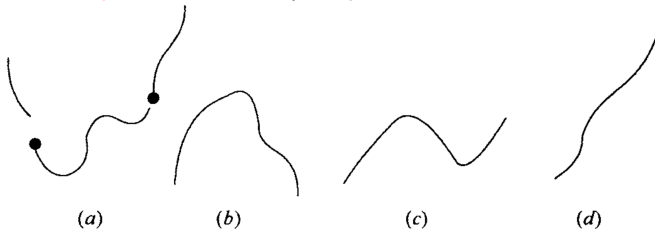
- 1 What is $\min\{f(\mathbf{x}) : \mathbf{x} \in S\}$?
- 2 What is $\max\{f(\mathbf{x}) : \mathbf{x} \in S\}$?

Quasiconvex Functions

Definition: Let S be a nonempty convex set in $\mathbb{R}^n$ and $f : S \rightarrow \mathbb{R}$. The function f is said to be **quasiconvex** if for each $\mathbf{x}_1, \mathbf{x}_2 \in S$ and $\lambda \in (0, 1)$:

$$f(\lambda \mathbf{x}_1 + (1 - \lambda) \mathbf{x}_2) \leq \max\{f(\mathbf{x}_1), f(\mathbf{x}_2)\}.$$

f is said to be **quasiconcave** if $-f$ is quasiconvex.

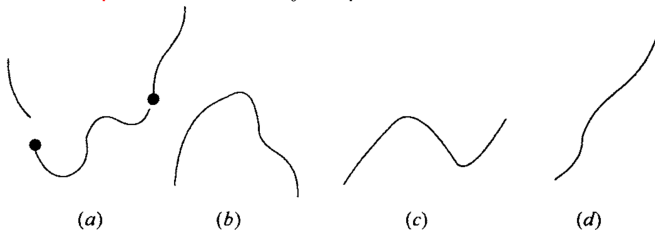


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Examples:

- (a) $f(x) = x^3$ is **quasiconvex**
- (b) $f(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$ is **quasiconcave**
- (c) Every convex function is also quasiconvex

Strictly Quasiconvex Functions

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Every local minimum to a minimization problem with a *strictly* quasiconvex objective and a convex feasible region is a global minimum.

Quasiconvex Maximization Has an Extreme Point Solution

Theorem

Let S be a nonempty compact polyhedral set in $\mathbb{R}^n$ and $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be quasiconvex and continuous on S . The problem $\max_{\mathbf{x} \in S} f(\mathbf{x})$ has an optimal solution that is an extreme point of S .